University of Cambridge Mathematical Tripos Part III

Analysis of PDEs

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Course description

This course serves as an introduction to the mathematical study of Partial Differential Equations (PDEs). The theory of PDEs is presently a major area of active research, and it goes back to the very birth of mathematical analysis in the 18th and 19th centuries. The subject lies at the crossroads of physics and many areas of pure and applied mathematics. The course will mostly focus on modern techniques, with emphasis on functional analytic techniques, relying on a priori estimates rather than explicit solutions. The course will primarily focus on approaches to linear, elliptic and evolutionary problems through weak formulations and energy estimates, with the prototypical examples being Laplace's equation, the heat equation, the transport equation, the wave equation and Schrödinger's equation. The following topics will be studied.

- *Introduction and Cauchy–Kovalevskaya theorem.* Definition and examples of PDEs; wellposed problems; classification; real analyticity; Cauchy–Kovalevskaya theorem.
- *Spaces of functions.* Hölder spaces; weak derivatives; Sobolev spaces; approximation by mollifiers; extension and trace theorems; Sobolev embeddings.
- *Elliptic boundary value problems.* Weak formulation; solvability via Lax–Milgram; Fredholm alternative for elliptic PDEs and its consequences on the spectrum; regularity of solutions via the difference quotient method.
- *Hyperbolic equations.* Notion of hyperbolicity for second-order linear operators; weak formulation of the initial boundary value problem; existence via Galerkin's method; uniqueness via the vector field method; finite speed of propagation; first-order hyperbolic PDEs and the method of characteristics.

Prerequisites

There are no specific pre-requisites beyond a standard undergraduate analysis background, in particular a familiarity with measure theory and integration. The course will be mostly self-contained and can be used as a first introductory course in PDEs for students wishing to continue with some specialised PDE Part III courses in the Lent and Easter terms.

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1 Introduction

1.1 Overview

An ordinary differential equation (ODE) is an equation of the form

$$F(x, u(x), u'(x), \cdots, u^{(k)}(x)) = 0, \qquad x \in I$$

where $u \in C^k(I)$ and $I \subset \mathbb{R}$ is an interval. Partial differential equations (PDEs) arise when u depends on several real variables, $u = u(x_1, \cdots, x_n)$ for $n \geq 2$. We work on a *domain*, i.e. an open, non-empty, connected subset $\mathcal{U} \subset \mathbb{R}^n$.

Definition Partial differential equation
Let $n \geq 2$. A PDE of order k is an equation

$$F(x, u(x), Du(x), \cdots, D^k u(x)) = 0, \qquad x \in \mathcal{U} \tag{1}$$

where $u \colon \mathcal{U} \to \mathbb{R}$ is the unknown and $F \colon \mathcal{U} \times \mathbb{R} \times \mathbb{R}^n \times \mathbb{R}^{n^2} \times \cdots \times \mathbb{R}^{n^k} \to \mathbb{R}$ a given function.

Remarks

- We can also have $u \in \mathbb{R}^p$ and $F \in \mathbb{R}^q$, yielding a system of PDEs.
- If $x_1 = t$, then we sometimes call the PDE *evolutionary*.

Definition Classical solution
A classical solution to the PDE (1) *is a function $u \in C^k(\mathcal{U})$ for which* (1) *holds pointwise on $\mathcal{U}$.*

Notation. $u_x = \partial_x u = \frac{\partial u}{\partial x}$, $Du = (\partial_{x_i} u)_i = \nabla u$, $D^2 u = (\partial^2_{x_i x_j} u)_{ij}$, etc.

Examples
Many partial differential equations arise from mathematical modelling in applied mathematics.

- SIR model

$$\dot{u}(t) = f(u, v)$$
$$\dot{v}(t) = g(u, v)$$

- Heat/diffusion equation

$$u_t = d\nabla u$$

  where d is the diffusion coefficient and $\Delta = \sum_i \partial^2_{x_i}$ is the Laplace operator.
- Reactor-diffusion equation

$$u_t = \nabla u + f(u, v)$$
$$v_t = \nabla v + g(u, v)$$

- Transport equation

$$u_t + \langle v, Du \rangle = 0$$

- Poisson equation

$$\nabla u = f(x)$$

  If $f = 0$, then this could describe an electrostatic potential. If $f \neq 0$, then this could describe a Newtonian gravitational potential.
- Euler equation and Navier–Stokes equations
- Maxwell's equations and the wave equation (hyperbolic PDEs)

$$\nabla \cdot \mathbf{E} = \rho, \qquad \nabla \cdot \mathbf{B} = 0, \qquad \partial_t = \nabla \times \mathbf{B} - \mathbf{J}, \qquad \partial_t \mathbf{B} = -\nabla \times \mathbf{E}$$

- Schrödinger equation, KdV equation (dispersive PDEs)

$$iu_t = -\nabla u + Vu$$

PDEs can also arise in pure mathematics:

- Cauchy–Riemann equations
- Ricci flow

$$\partial_t g = -2\,\mathrm{Ric}_g$$

All of the equations in the examples above need additional boundary information to solve, e.g. $u|_{\partial\mathcal{U}}$ for the Laplace equation. Such boundary information is the *data* of the problem. There is no single theory for PDEs — one instead exploits structure (elliptic/hyperbolic/parabolic) and other properties (scaling symmetry, invariances, types of data, etc.). In this course, we take a modern approach to PDEs in the sense that we study existence, uniqueness and other qualitative properties instead of finding explicit representation formulae for the solutions.

1.2 Classes of PDEs

Notation. In this course, we will use *multi-index notation*. For a *multi-index* $\alpha = (\alpha_1, \cdots, \alpha_n) \in \mathbb{N}^n$ of *order* $|\alpha| = \alpha_1 + \cdots + \alpha_n$, we define

$$D^\alpha f(x) := \frac{\partial^{|\alpha|}}{\partial_{x^1}^{\alpha_1} \cdots \partial_{x_n}^{\alpha_n}} f(x)$$

Definition Linear
We say that (1) *is linear if F is linear in u and its derivatives, i.e.*

$$\sum_{\alpha\colon |\alpha|\le k} a_\alpha(x) D^\alpha u(x) = f(x)$$

If $f \equiv 0$, we say that the PDE is homogeneous.

Definition Semilinear
We say that (1) *is semilinear if the highest-order derivatives appear linearly with coefficients depending only on x, i.e.*

$$\sum_{\alpha\colon |\alpha|=k} a_\alpha(x) D^\alpha u(x) + a_0(x, u, \cdots, D^{k-1}u) = 0$$

Definition Quasilinear
We say that (1) *is quasilinear if the highest-order derivatives appear linearly with coefficients depending on x and the lower-order derivatives, i.e.*

$$\sum_{\alpha\colon |\alpha|=k} a_\alpha(x, u, Du, \cdots, D^{k-1}u) D^\alpha u(x) + a_0(x, u, \cdots, D^{k-1}u) = 0$$

Definition Fully nonlinear
We say that (1) *is fully nonlinear if it is not quasilinear.*

1.3 Lessons from ODEs

Consider the ODE

$$\begin{cases} u'(t) = f(t, u(t)) \\ u(t_0) = f_0 \end{cases} \tag{2}$$

The goal is to show existence and uniqueness for solutions depending on the regularity of f. There are roughly three key theorems that establish this.

Theorem 1.1 Cauchy–Kovalevskaya for ODEs
Suppose f is a real analytic function on a neighbourhood of (t_0, u_0). Then there exists a unique real analytic solution of (2) *defined on a neighbourhood of t_0.*

Theorem 1.2 Cauchy–Lipschitz / Picard–Lindelöf
Suppose f is continuous in both variables and Lipschitz in the second variable u. Then there exists a unique C^1 solution to (2) *defined on a neighbourhood of t_0.*

Remark. By considering the ODE $u'(t) = u(t)^2, u_0 > 0$, the solution to this ODE given by the above theorem is only guaranteed to hold *locally*.

Theorem 1.3 Cauchy–Peano
Suppose f is continuous. Then (2) *admits at least one C^1 solution.*

Remark. By considering the ODE $u'(t) = \sqrt{u(t)}, u_0 = 0$, we indeed see that solutions given by the above theorem need not be unique.